

Session 1

Calibration, Traceability & Metrological Standards I



Chair: Marija Cundeva-Blajer

Thursday 17 September 2026, 08:55–10:30 • Room T2:C3:340

08:55–09:30 **One Bit to Useful Knowledge: Rethinking Data Acquisition for Low-Power and High-Speed Measurement Systems** **KEYNOTE**

Prof. Paolo Carbone

09:30–09:45 **Development of System for Disseminating AC/DC Voltage Difference Using Ranges of Transfer Standard and Investigating Stability**

Valentyn Isaiev

09:45–10:00 **MetroMag: a European infrastructure for low magnetic field metrology**

Marco Coïsson

10:00–10:15 **Franco Cabiati, electrical impedance metrology, and IMEKO**

Luca Callegaro

10:15–10:30 **Exploring calibration schemes for electric impedance measurements beyond the short-open-load**

Alessandro Cultrera

Session 2

Calibration, Traceability & Metrological Standards II



Chair: Luca Callegaro

Thursday 17 September 2026, 11:00–13:00 • Room T2:C3:340

- 11:00–11:30** **Major European Research Projects in Electrical Metrology** **KEYNOTE**
Prof. Jiří Tesař
- 11:30–11:45** **High-Temperature Superconductor Josephson Junction Arrays Coupled to Microwave of Different Polarizations**
Changhao Zhao
- 11:45–12:00** **MEMS-based Long-haul Optical Link Asymmetry Estimation for Atomic Clock Comparison**
Michal Spacek
- 12:00–12:15** **A Model for Secure Exchange of Digital Calibration Certificates**
Ahmed Khan Leghari
- 12:15–12:30** **Metrological Traceability in IoT Sensor Networks: A Microservice-based Framework for Automated Calibration Management**
Marco Sento
- 12:30–12:45** **Implementation of Monte Carlo Simulation for measurement uncertainty estimation in electrical power and energy reference standards calibration**
Kiril Demerdziev
- 12:45–13:00** **The Use of Thermography for Non-Destructive Testing**
Antonín Krňoul

Session 3

Uncertainty Evaluation & Electrical/Magnetic Instrumentation



Chair: Paolo Carbone

Thursday 17 September 2026, 14:00–15:30 • Room T2:C3:340

14:00–14:30 **Measuring the Performance of C-UAS Systems: How Hard Can It Be?** **KEYNOTE**

Dr. Cristian Coman

14:30–14:45 **GUM-compliant uncertainty evaluation in measurements based on ML regression or classification models**

Leopoldo Angrisani

14:45–15:00 **Quality Infrastructure for Resilient Electrical Sector - Ensuring Confident Measurement Results in Calibration of High Currents Instruments**

Marija Cundeva-Blajer

15:00–15:15 **Development of the Digital Twin for a Digital Impedance Ratio Bridge**

Ana Lucia Medina Rodriguez

15:15–15:30 **Accuracy Analysis of THD Measurement of Power Signals by Means of a Low-cost Oscilloscope**

Gabriel Gasparesc

Session 4

Electrical & Magnetic Instrumentation



Chair: Cesare Svelto

Thursday 17 September 2026, 16:00–17:15 • Room T2:C3:340

16:00–16:15 The primary Tesla Standard at PTB

Franziska Weickert

16:15–16:30 A traceable calibration of low-field magnetic coil systems in noisy industrial environments

Michal Ulvr

16:30–16:45 Model-based magnetization measurement in Fluxgate sensors

Jiri Maier

16:45–17:00 Estimation of torquer rod magnetic moment with fluxmetric method and compact sensing coils

Michal Janosek

17:00–17:15 Compact modular cryogenic dielectric resonator for surface impedance measurements in high magnetic fields

Kostiantyn Torokhtii

Session 5

ADC/DAC, Waveform & Data Conversion Metrology



Chair: Jan Holub

Friday 18 September 2026, 08:30–10:00 • Room T2:C3:340

08:30–09:00 Artificially indeterminate: measuring quantity, cost and impact of Artificial Intelligence **KEYNOTE**

Dr. Michael Street

09:00–09:15 A systematic aliasing suppression architecture for real-time FI-DAC based on linear-phase complementary filter and oversampling

Yang Shen

09:15–09:30 Parameter estimation of complex-valued sinewaves by a frequency-domain linearized sine-fit algorithm

Dominique Dallet

09:30–09:45 Benchmarks for IIR Model Identification Based on Binary Measurements

Paolo Carbone

09:45–10:00 A Digital Voltage Ramp Generator with Integrating Compensation

Jianting Zhao

Session 6

Advanced Sensors & Signal Processing I



Chair: Pedro M. Ramos

Friday 18 September 2026, 10:30–12:15 • Room T2:C3:340

10:30–11:00 **Trust by Design in Digital Metrology: From Machine-Readable Documents to Privacy-Preserving Computation** **KEYNOTE**

Prof. Kruno Miličević

11:00–11:15 **Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds**

Martin Maxa

11:15–11:30 **Lock-in detection algorithm for signals modulated with irregular intervals**

Shingo Tamaru

11:30–11:45 **A Wireless Acoustic Monitoring System for Drone Localization Based on MEMS Microphones and LoRa Communication**

Kamil Herman

11:45–12:00 **Comparative Analysis of Search Algorithms for Automatic L-type Matching Network**

Joao Pedro Torres S. Andrade

12:00–12:15 **Airborne Navigation During GNSS Outages Using Tightly Coupled IMU/DVB-T2 Fusion**

Michal Koldinsky

Session 7

Advanced Sensors & Measurement Systems II



Chair: Luca De Vito

Friday 18 September 2026, 13:15–14:30 • Room T2:C3:340

13:15–13:30 Microcontroller Based Software Defined Instruments as an Economical Substitute of Virtual Measurement Instruments

Jakub Svatos

13:30–13:45 Exploring a new indirect approach to latency measurement

Jakub Turinsky

13:45–14:00 Detection of Mercury Ion in Water Samples Using a Low-Tech EPS-Coated Interdigitated Capacitive Sensor through S-Parameter Analysis

Henrique Dantas Silva

14:00–14:15 Development and Experimental Evaluation of a Three-Lead ECG Holter System

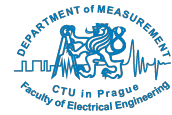
Ondrej Kovac

14:15–14:30 Multiparametric metrological characterization of an inductive sensing element for mechanical wear estimation

Davide Ceschini

Session 8

Applications in Energy Systems, Emerging Technologies & Materials Characterization



Chair: Gabriel Gasparesc

Friday 18 September 2026, 15:00–16:45 • Room T2:C3:340

- 15:00–15:15** **EUROLAB: Empowering Digital and Sustainable Labs and other CABs** KEYNOTE
Laura Martin
- 15:15–15:30** **Detection of Physical Layer Anomalies and Eavesdropping in Real World Data Cables**
Luca De Vito
- 15:30–15:45** **Sustainability-oriented optimization of a sensing board for applications as a cyber-physical measurement system (CPMS)**
Leopoldo Angrisani
- 15:45–16:00** **Vector Fitting-Based Feature Extraction from Electrochemical Impedance Spectroscopy for SOC and SOH Estimation**
Pedro Ramos
- 16:00–16:15** **Measurement of the characteristics of EDL and hybrid supercapacitors versus temperature**
Ornella Conti
- 16:15–16:30** **Non-Destructive Microwave Dielectric Characterization of Leather for Automotive Applications**
Kostiantyn Torokhtii
- 16:30–16:45** **Measurements of Wetsuits Thermal Properties at 100 kPa and 600 kPa Absolute Pressure**
Cesare Svelto